Trigonometric Functions and Derivatives

MATH 145 – Applied Calculus

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Radians

Generally, in real life, and in your high school geometry classes, you might have used *degrees* to measure the size of angles. In calculus, we tend to use the *radian* instead.

Radian

The radian measure of the size of an angle is the length of the arc of a circle of radius $r = 1$ swept out by the angle, with positive values corresponding to counterclockwise rotation and negative values corresponding to clockwise rotation.

The circumference of the full unit circle is $C = 2\pi(1) = 2\pi$, so $360° = 2\pi$ radians.

In general, for a circle of radius r, the arc length swept out by an angle of θ radians is $r\theta$.

Radians

Using the fact that $360° = 2\pi$ radians, we can convert from degrees to radians by multiplying the degrees by $\frac{\pi}{180}$, and similarly, we can convert from radians to degrees by multiplying the radians by $\frac{180}{\pi}$.

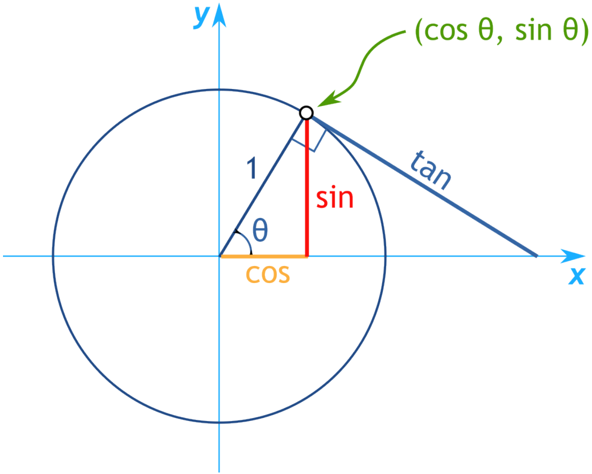
1. Convert $115°$ to radians.
2. Convert $\frac{13\pi}{16}$ radians to degrees.

Sine and Cosine

For an angle θ, $\cos(\theta)$ and $\sin(\theta)$ are the (x, y) coordinates of the intersection of the terminal side of the central angle with radian measure θ and initial side on the positive x–axis.

In other words, the coordinates of the point θ units around the unit circle are $(\cos(\theta), \sin(\theta))$.

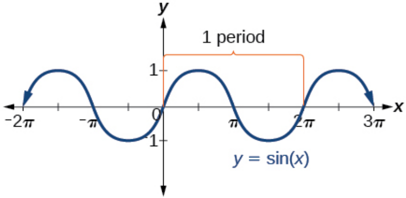
Trigonometric Functions

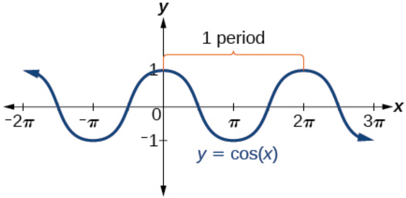


It follows from this that $\cos^2(\theta) + \sin^2(\theta) = 1$

Graph of Sine and Cosine

As $-\infty < t < \infty$, $\sin(t)$ and $\cos(t)$ trace out the graph of a function:





These functions are **periodic** with period 2π.

Periodic Functions

$\sin(t)$ and $\cos(t)$ (and combinations of them) can be used to model any periodic phenomenon using stretches and shifts of functions, e.g., $\cos(t) = \sin(t + \frac{\pi}{2})$.

$$f(x) = A\sin(Bt + C) + D$$

- Amplitude: A
- Period: $P = \dfrac{2\pi}{B} \Longleftrightarrow B = \dfrac{2\pi}{P}$
- Phase Shift: C
- Shift: D

Example: Constructing Periodic Functions

Example: In a port city, high tide is at 9.9 feet, and low tide is 0.1 feet. High tide is at midnight and low tide is at noon. Find a formula for the height of the tide as a function of hours since midnight.

Other Trigonometric Functions

Other well-known trigonometric functions can be defined in terms of $\sin(x)$ and $\cos(x)$:

- Tangent: $\tan(x) = \dfrac{\sin(x)}{\cos(x)}$
- Secant: $\sec(x) = \dfrac{1}{\cos(x)}$
- Cosecant: $\csc(x) = \dfrac{1}{\sin(x)}$
- Cotangent: $\cot(x) = \dfrac{\cos(x)}{\sin(x)}$

Derivatives of Trigonometric Functions

$$\frac{d}{dx}\sin(x) = \cos(x) \qquad \text{and} \qquad \frac{d}{dx}\cos(x) = -\sin(x)$$

Why? Use the definition of the derivative, the angle sum formula ($\sin(x+h) = \sin(x)\cos(h) + \cos(x)\sin(h)$) and some facts about limits:

$$\begin{aligned}
\frac{d}{dx}\sin(x) &= \lim_{h\to 0} \frac{\sin(x+h) - \sin(x)}{h} \\
&= \lim_{h\to 0} \frac{\sin(x)\cos(h) + \cos(x)\sin(h) - \sin(x)}{h} \\
&= \sin(x)\left(\lim_{h\to 0} \frac{\cos(h) - 1}{h}\right) + \cos(x)\left(\lim_{h\to 0} \frac{\sin(h)}{h}\right) \\
&= \sin(x)(0) + \cos(x)(1) \\
&= \cos(x)
\end{aligned}$$

Examples: Derivatives of Trigonometric Functions

1. Find $\dfrac{d}{dx}\tan(x)$.
2. Find $\dfrac{d}{dx}\sec(x)$.
3. Find $\dfrac{d}{dx}\csc(x)$.
4. Find $\dfrac{d}{dx}\cot(x)$.

Examples: Derivatives of Trigonometric Functions

5. Find $\dfrac{d}{dx}\sin^2(x)$.

6. Find $\dfrac{d}{dx}\cos(x^3 + x)$.

7. Find $\dfrac{d}{dx}\sqrt{x}\sin(3x^2 + 2)$.